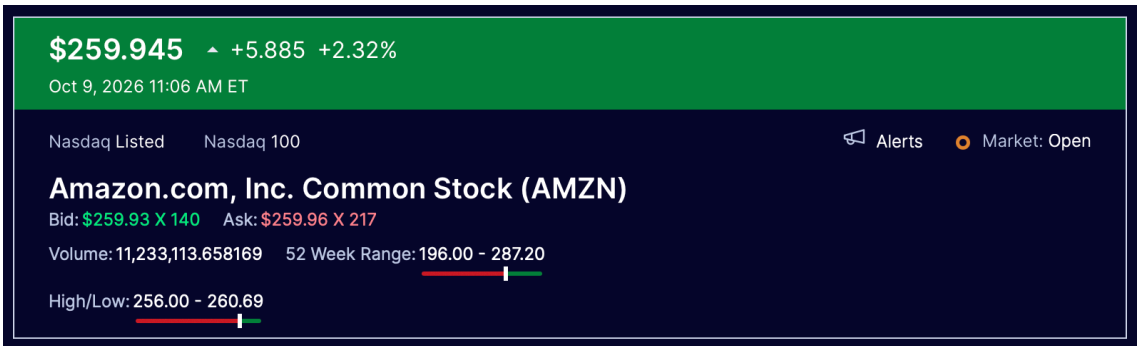
Mandatory Assignment



1 Amazon volatility surface

On October 9, 2026, Amazon.com (ticker: AMZN) trades at USD 260.

The Excel-file `AMZN_options.xlsx` shows prices of a range of standard European call options. There are five expiries: 14, 31, 93, 183, and 365 calendar days. For each expiry, there are 13 strikes. In this assignment, you should use a constant interest rate of $r = 5\%$ and a dividend yield of zero (AMZN does not pay dividends).

Questions

1. For each option, compute the Black-Scholes implied volatility
2. For each option maturity, plot the implied volatility smile; i.e. implied volatility as a function of $\log(K/F_t)$, where $F_t = S_t e^{r(T-t)}$ is the forward price
3. Characterize the implied volatility patterns
4. For each option maturity, plot also the Black-Scholes delta and vega as a function of $\log(K/F_t)$

2 Calibrating the Heston model

Suppose you work for an investment bank that trades in exotic derivatives on AMZN. For the purpose of pricing such derivatives, you need a model. You choose the Heston model. Calibrate the Heston model to the AMZN implied volatility surface. Specifically, instead of calibrating directly to implied volatilities, calibrate to option prices scaled by their Black-Scholes vegas. Basically, this way you convert option pricing errors in terms of prices into option pricing errors in terms of implied volatilities, through a linear approximation

- Let $\widehat{P}_t(K,T)$ and $P_t(K,T)$ denote fitted and actual option prices, respectively
- Let $\widehat{\sigma}_t^{BS}(K,T)$ and $\sigma_t^{BS}(K,T)$ denote the corresponding BS implied volatilities
- Let $\mathcal{V}_t^{BS}(K,T)$ denote the BS vegas computed at the true option prices
- Then

$$\widehat{P} \approx P + \mathcal{V}^{BS}\left(\widehat{\sigma}^{BS} - \sigma^{BS}\right) \Leftrightarrow \frac{\widehat{P} - P}{\mathcal{V}^{BS}} \approx \widehat{\sigma}^{BS} - \sigma^{BS}.$$

The advantage is that you avoid having to compute implied volatilities from prices during the calibration, which would slow down the calibration considerably.

Use the following parameter values as starting values for the calibration: $\kappa = 3$, $\theta = 0.10$, $\rho = -0.20$, $\sigma = 1$, and $v_0 = 0.10$.

Questions

1. Report and comment on the parameter estimates
2. For each option maturity, plot the fitted and actual implied volatility smile; i.e. implied volatility as a function of $\log(K/F_t)$
3. Comment on the fit of the model. Does it correspond to your expectations?

3 Simulating the Heston model

To avoid having an error from your calibration carry over to the rest of the assignment, use the following parameter values for the Heston model: $\kappa = 4$, $\theta = 0.12$, $\rho = -0.15$, $\sigma = 0.80$,

and $v_0 = 0.06$. When simulating the Heston model use the "full truncation" simulation scheme discussed in class, and use a time step of $\Delta t = 1/252$.

Questions

1. Simulate 100,000 paths of S_t and v_t out to 6 months using antithetic variates (i.e., 50,000 paths and 50,000 antithetic paths). Use these paths throughout the rest of the assignment. Plot the simulated conditional distribution of $\log(S_T)$ vs. the normal distribution with same mean and variance. Plot also the simulated conditional distribution of v_T.

2. Report the mean, median, skewness and kurtosis of the simulated distributions of $\log(S_T)$ and v_T. Comment on the results.

3. Verify the simulation design by pricing a 6-month European put with strike $K = 270$ on the simulated paths. Report the price and its 95 percent confidence interval, and compare with the quasi-analytical price computed from the characteristic function. Is the analytical price inside the confidence interval? If not, what is the likely reason?

4 Barrier options

A client wants to buy a 6-month down-and-out put option on AMZN with strike $K = 250$ and barrier $S_b = 150$. The barrier is monitored daily at the closing price, and the option is knocked out if $S_t \leq S_b$ on any monitoring date.

Questions

1. Price the down-and-out put on the paths from Section 3. Report the price and the 95 percent confidence interval.

2. Price the corresponding down-and-in put on the same paths. Explain why the sum of the two barrier prices equals the simulated price of the European put with the same strike and maturity exactly, and not only up to simulation error.

3. Use the European put as a control variate for the down-and-out put. Report the estimated coefficient b, the control variate price and its confidence interval, and the

reduction in the standard error relative to question 1. Explain why the European put is a good control.

4. Price the down-and-out put again with weekly monitoring (every fifth time step). Explain the direction of the change, and what it implies for a continuously monitored barrier.

5 American options

A client wants to buy a 6-month American put option on AMZN with strike $K = 270$. The option can be exercised daily at the closing price.

Questions

1. Price the American option with the LSM algorithm of Longstaff and Schwartz (2001) on the paths from Section 3. Consider the following five sets of basis functions:

$$\{1, S_t/K, (S_t/K)^2, v_t, v_t^2\}$$
$$\{1, S_t/K, (S_t/K)^2, (S_t/K)^3, v_t, v_t^2, v_t^3\}$$
$$\{1, S_t/K, (S_t/K)^2, (S_t/K)^3, v_t, v_t^2, v_t^3, v_t S_t/K\}$$
$$\{1, S_t/K, (S_t/K)^2, (S_t/K)^3, v_t, v_t^2, v_t^3, v_t S_t/K, v_t^2 S_t/K, v_t (S_t/K)^2\}$$
$$\{1, S_t/K, (S_t/K)^2, (S_t/K)^3, (S_t/K)^4, v_t, v_t^2, v_t^3, v_t^4, v_t S_t/K, v_t^2 S_t/K, v_t (S_t/K)^2\}\,.$$

   For each set of basis functions, provide both the price and the associated 95 percent confidence interval (do not change the paths when changing the basis functions). Comment on the results.

2. Compare with the European put with the same strike and maturity, priced by simulation on the same paths (this is the price from Section 3, question 3). Report the early exercise premium, i.e. the difference between the American and the European price, and explain why it is positive.